

Product Data Sheet: David CG Thermostat

February 2011



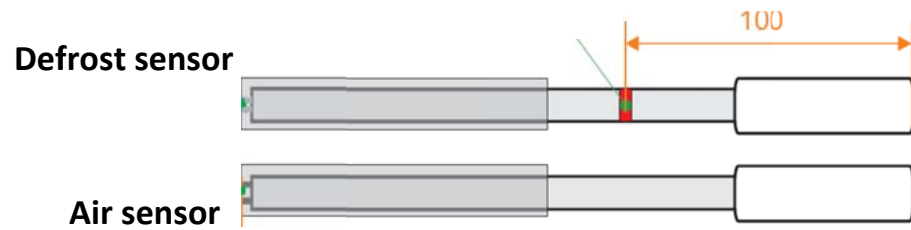
General	
David thermostat – HNK-P0003668-A	
Supplier code article-number: 6114 553	
Country of origin: Germany	

Dimensions	
Length	81 mm
Width	36 mm
Depth	65,29 mm

Details / Information	
Power Supply	230V AC +10/-15%, 50/60Hz
Rated power	1,5 VA
Display	3 digit LED display with sign -199 to 199 with decimal point, six status LED's
Case – Material	Plastic terminal
Operating conditions	-10T50°C, humidity <95%rH (with coating of electronic)
Storage conditions	-20T70°C, humidity <95%rH (with coating of electronic)
Measurement range	-50T90°C, resolution 0,1°C

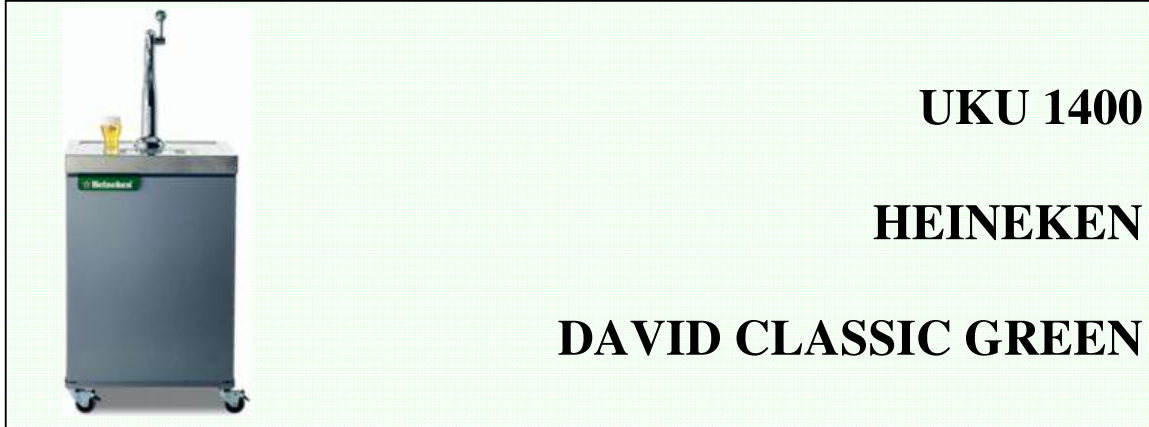
Product Data Sheet: David Sensors (air + defrost)

February 2011



General	
David CG air sensor + defrost sensor – HNK-P0003668-A	
Supplier code article-number: 9406 199 (6942 795 air sensor + 6942 463 defrost sensor)	
Country of origin: Germany	

Details / Information	
Colour of Cable defrost sensor	White
Colour of Cable air sensor	Black
Length	460 mm



TECHNICAL INFORMATION

CAREL EASY CONTROLLER PYLB1K050R [6114 553-00]

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UKU 1400 – HEINEKEN – DAVID CLASSIC GREEN

TECHNICAL INFORMATION

CAREL EASY CONTROLLER PYLB1K050R [6114 553-00]

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CAREL EASY CONTROLLER PYLB1K050R [6114 553-00]

1. Scope

For the new UKU1400 “Heineken David Green Classic” an electronic control will be built in. Because of the good experience with Carel-easy-controllers Liebherr decided to use this type.

Requirements:

- 1 relays for compressor, 2 HP
- 1 changeover-relay for fans / light
- 2 sensor inputs (control sensor / defrost sensor)
- 1 digital input for door-contact

Because of these requirements the Carel-type PJEZY0H00 has been chosen.

To adapt this type to the Liebherr-specific requirements Carel has to optimize the controller's software. Therefore Carel created a new code:

PZLBY0H001	-single pack
PZLB1K050R	-multiple pack 50pcs.

New Liebherr article number: 6114 553-00



2. Technical Specification

2.1 General

Power supply:	230V AC +10/-15%, 50/60Hz
Rated power:	1,5VA
Relay outputs:	2HP relay compressor / 8A relay fans/light
Probe type:	NTC 10kΩ at 25°C
Display:	3 digit LED display with sign -199 to 199 with decimal point, six status LED's
Operating conditions:	-10T50°C, humidity <95%rH (with coating of electronic)
Storage conditions:	-20T70°C, humidity <95%rH (with coating of electronic)
Measurement range:	-50T90°C, resolution 0,1°C
Case:	Plastic terminal, 81x36x65mm
Protection against el. shock:	Class II
Environmental pollution:	Normal
PTI of insulating materials:	250V
Resistance to heat and fire:	Category D (UL94-VO)

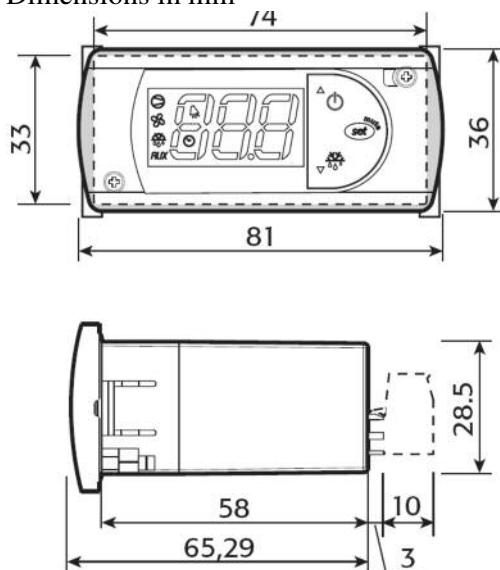
2.2 Electromagnetic compability

The “easy” series is compliant with the EU standards on electromagnetic compatibility:

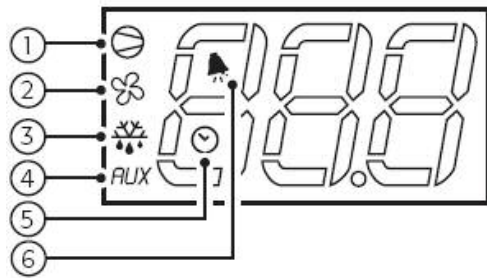
- appliances for domestic environments EN55014-2 and EN55014-1;
- residential, commercial and light industrial environments EN50082-1 and EN50081-1;
- industrial environments EN50082-2 and EN50082-1;
- as regards safety the device is compliant with standards EN60730-1 and EN60730-2.

2.3 Dimensions

Dimensions in mm



2.4 Description of the display

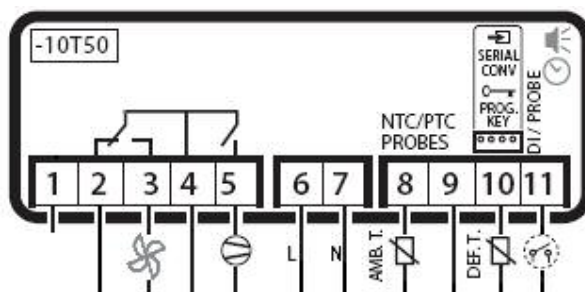


The display has 3 digits with decimal point and 6 symbols (compressor, fans, defrost, AUX, alarm and clock).

Key:

1	Compressor: comes on when the compressor is started. Flashes when the activation of the compressor is delayed by protection times or by a procedure in progress. Flashes with 2 cycles and a pause when the continuous cycle is enabled.
2	Fan: comes on when the evaporator fans are started. Flashes when the activation of the evaporator fans is prevented due to external disabling or procedures in progress
3	Defrost: comes on when the defrost is started. Flashes when the activation of the defrost is prevented due to external disabling or procedures in progress
4	AUX: comes on when the auxiliary output is activated
5	Clock: comes on if fitted, enabled (parameter "tEn") and at least one time band has been set. At startup comes on for a few seconds to indicate that the option is available.
6	Alarm: comes on when an alarm situation occurs.

2.9 Electrical connection



- 1 ...free
- 2 ...changeover-relays light
- 3 ...changeover-relays fan
- 4 ...phase conductor
- 5 ...compressor-relay
- 6 ...phase conductor
- 7 ...neutral conductor
- 8 ...control sensor
- 9 ...common
- 10 ...defrost sensor
- 11 ...door-contact (digital input)

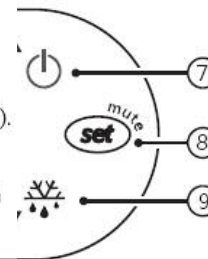
2.7 Description of the buttons

Information:

Pressing each button briefly, the display shows the message relating to the associated function.

Key:

7	UP/ON OFF In normal operation: if pressed for more than 3 seconds changes the status of the controller (ON/OFF). if pressed together with DOWN for more than 3 seconds, activates /deactivates the continuous cycle (when pressed the display shows "cc"). In parameter setting mode: moves from one parameter to the next; increases the value associated with the parameter. In set point display mode: increases the value of the set point
8	SET/MUTE In normal operation: mutes the audible alarm (buzzer); if pressed for more than 1 second displays the set point; if pressed for more than 3 seconds, accesses the menu for setting the parameters; if pressed together with DOWN when switching the instrument on, activates the procedure for setting the default parameters ("CF" on the display). In parameter setting mode: alternates the display of the value of each parameter with the name; if pressed for more than 3 seconds saves the changes. In set point display mode: if pressed saves the changes.
9	DOWN/DEFROST In normal operation: if pressed for more than 3 seconds, activates /deactivates the manual defrost; if pressed together with UP for more than 3 seconds, activates/deactivates the continuous cycle (when pressed the display shows "cc"), if pressed when switching the instrument on displays the software version if pressed together with SET when switching the instrument on, activates the procedure for setting the default parameters ("CF" on the display). In parameter setting mode: moves from one parameter to the previous; decreases the value associated with the parameter. In set point display mode: decreases the value of the set point.

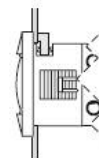
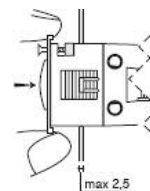


2.8 Assembly

Panel installation from the front using screws (Fig. 3.b):

The thickness of the fastening panel must not exceed 3 mm;

- remove the front frame and make sure that the two catches are in place (these must not protrude from the outline of the drilling template). If necessary, unscrew the two screws. Do not unscrew excessively, the screws must not be detached from the front panel (phase 1);
- insert the instrument in the opening in the panel and hold it in position by the centre of the front panel (phase 1);
- using the screwdriver, tighten the bottom screw 90°, the catch must come out of its slot and click onto the panel, then tighten until the front panel is secure; Do not over-tighten, when the front panel is secured blocks simply make another ½ turn to compress the gasket; If the catch does not click onto the panel, unscrew the screw, applying pressure at the same time with the screwdriver so that the catch moves back. Do not unscrew too much, the head of the screw must not be raised from the surface of the front panel (phase 2);
- repeat the same operation for the top screw (phase 2);
- the two screws must be tightened with the same pressure, so as to not leave one corner higher than the other. Do not tighten excessively, when the front panel is secure simply tighten a further ½ turn to compress the gasket (phase 2);
- apply the front frame (phase 3).



Dismantling using the screws from the front:

- unclip the front frame;
- unscrew the bottom screw, at the moment the front panel detaches from the panel keep pressure on the screw and unscrew a further 90° to make the catch go back into its slot;
- repeat for the top screw;
- remove the instrument from panel, keeping it horizontal

3. Functions and Parameters

3.1 Modifying the parameters

The operating parameters, set using the front keypad, are divided into two families: frequent (type F) and configuration (type C). Access to the latter type is protected by a password that prevents random modifications or entry by unauthorised persons.

Accessing the type F parameters:

- press the SET button for more than 3 s (in the event of alarms, first mute the buzzer) the display shows the code of parameter 'PS' (password);
- use the UP and DOWN buttons to scroll the names of the parameters. The scrolling of the parameters is accompanied by an icon on the display that represents (where possible) the category the parameter belongs to;
- press SET to display the value associated with the parameter;
- increase or decrease the value using the UP or DOWN button respectively;
- press SET to temporarily save the new value and return to the display of the parameter;
- repeat the operations if necessary
- press the SET button for more than 3 seconds to permanently save the parameters and exit the parameter setting procedure.

Accessing the type C parameters:

- press the SET button for more than 3 s (in the event of alarms, first mute the buzzer) the display shows the code of parameter 'PS' (password);
- press the SET button to enter the password;
- use the UP and DOWN buttons to scroll the numbers until displaying "22" (password to access the parameters);
- press the SET button to confirm the password;
- use the UP and DOWN buttons to scroll the names of the parameters. The scrolling of the parameters is accompanied by an icon on the display that represents (where possible) the category the parameter belongs to;
- press SET to display the value associated with the parameter;
- increase or decrease the value using the UP or DOWN button respectively;
- press SET to temporarily save the new value and return to the display of the parameter;
- repeat the operations if necessary;
- press the SET button for more than 3 seconds to permanently save the parameters and exit the parameter setting procedure.

Warning:

- if no button is pressed for more than 60 seconds, all the changes made to the parameters, temporarily saved in the RAM, will be cancelled and the previous settings restored.
- The parameters relating to the clock are not restored, as these parameters are saved when entering. The instrument then returns to normal operation.
- if power is disconnected from the instrument before saving the values (pressing the button SET for 3 seconds) all the changes made to the parameters and temporarily saved will be lost.

The parameters, as well as being divided by type, are grouped into logical categories identified by the initial letters or symbols on the display, as described below.

Category	Initial	Icon
Probe parameters	/	-
Control parameters	r	-
Compressor parameters	c	
Defrost parameters	d	
Alarm parameters	A	
Fan parameters	F	
AUX output configuration parameters	HI	AUX
RTC parameters	-	

3.2 Description of the UKU1400 relevant parameters

/2 Probe measurement stability

This defines the coefficient used to stabilise the temperature measurement. Low values assigned to this parameter offer the prompt response of the sensor to variations in temperature; the reading is however more sensitive to disturbance. High values slow down the response but guarantee greater immunity to disturbance, that is, a more stable and more precise reading.

The parameter filters the minimum variations in the temperature measurements and, at the same time, considers the mean value of the readings.

/C1 Probe 1 offset

This parameter is used to correct the temperature measured by the probe 1 with an offset in tenths of a degree. The value assigned to this parameter is in fact added to (positive value) or subtracted from (negative value) the temperature measured by probe 1.

/4 Select probe displayed

Used to choose whether to display the temperature read by the control probe (Probe 1) of Probe 2 or Probe 3.

Parameter /4 selects the probe shown on the display, all the other display and control modes remain unchanged.

/5 Select °C/°F

Defines the unit of measure used for the control and the display.

0= degrees centigrade

1= degrees Fahrenheit.

/6 Disable decimal point

Used to enable or disable the temperature display with the resolution to the tenths of a degree between -20 and +20.

0= the temperature values are displayed to the tenth of a degree, between -20 and +20;

1= all the temperature values are displayed without the tenths.

Note: the decimal point is only disabled in relation to the reading shown on the main and remote displays, while the parameters are always set to the tenth of a degree.

St User set point

Represents the temperature set point defined by the user.

r1 Minimum set point allowed

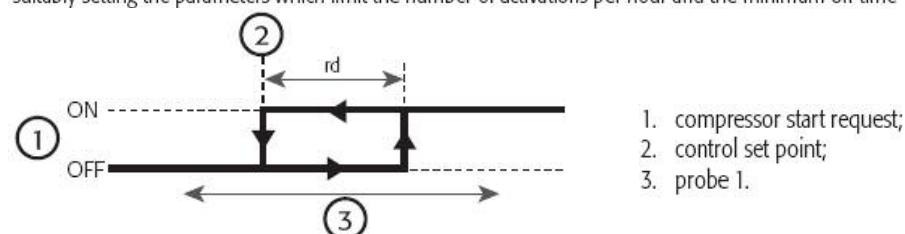
Defines the minimum set point that can be entered on the user interface using the UP and DOWN buttons or via supervisor.

r2 Maximum set point allowed

Defines the maximum set point that can be entered on the user interface using the UP and DOWN buttons or via supervisor.

rd Differential

Establishes the value of the differential, or hysteresis, used for temperature control. A narrow, that is numerically small differential, guarantees an ambient temperature which stays close to the set point, yet with frequent starts and stops of the compressor. The life of the compressor can still be protected by suitably setting the parameters which limit the number of activations per hour and the minimum off time



c4 Duty setting time

c4= 0 in the event of an ambient probe fault, the compressor will be always off;

c4= 100 the compressor will remain always on; and the 15 minute off time is not considered.

0<c4< 100 at the moment a probe error (E0) is detected, the duty setting cycle starts from the current status of the compressor (that is, the status of the compressor is maintained) and the time already elapsed is ignored.

The time settings for compressor C1, C2, C3 are in any case always observed.

If the control probe error occurs when the control is in defrost mode, the function is not stopped but rather is completed as required; in addition, the defrost can still be performed when called. During the defrost, the duty setting timer is not stopped but rather keeps counting.

If on the other hand the continuous cycle is operating, the control instantly exits this status and activates duty setting (also for safety reasons, given that during the continuous cycle the low temperature alarm is disabled). To re-activate continuous cycle operation, the ambient probe fault must be resolved.

If the probe error is no longer present, the unit returns to normal operation. The compressor again takes on the logic of the control, respecting the times C1, C2, C3.

DUTY SETTING is a safety function that ensures the operation of the compressor even in the event of a control probe fault. If alarm E0 occurs this parameter is used to ensure the operation of the compressor until the fault is resolved. Essentially, the compressor, not being able to be activated according to temperature, works cyclically with an ON time equal to the value assigned to parameter c4 (in minutes) and a fixed OFF time of 15 minutes.

AL Absolute or relative temperature for low temperature alarm

Used to determine the activation threshold for the low temperature alarm.

A0 Alarm and fan differential

A0≤0 AL expressed as an absolute (Fig. 4.c)

A0>0 AL expressed relatively to the set point (St or St + r4 in night-time operation) (Fig. 4.d)

St User set point

Represents the temperature set point defined by the user.

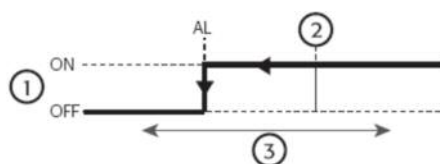


Fig. 4.c

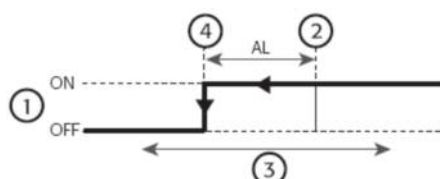


Fig. 4.d

Key to Figs. 4.c to 4.d

- 1. continuous cycle operation;
- 2. control set point;
- 3. probe 1;
- 4. AL - control set point.

c0 start delay after power-on

Represents the delay in starting the compressor and the evaporator fans from instrument power-on.

c2 minimum compressor off time

Establishes the time (in minutes) that the compressor remains off. The compressor is not started again until the minimum time set for 'c2' has elapsed since the last shutdown.

This parameter is useful for balancing the pressure after the compressor stops in systems with hermetic and capillary compressors. (Fig. 4.g)

c3 minimum compressor on time

Establishes the time (in minutes) that the compressor runs for. The compressor is not stopped until it has operated for a time equal to the minimum set value 'c3'. (Fig. 4.h)

dl Defrost interval

The defrost is performed periodically at a frequency of every dl (in hours or minutes, see parameter dC). The interval dl starts counting from the previous end defrost. If the time is equal to 0 (dl=0), the defrost is not performed, except when activated from the keypad, by digital input, from the supervisor or by time band.

During defrosts the temperature alarms are disabled.

Note: not referring to parameter dl

dt end defrost temperature set point / defrost temperature threshold with temp. control

For units fitted with end defrost probe, this parameter allows an evaporator temperature to be set at which the defrost is stopped (the evaporator temperature is measured by the defrost probe). The end defrost temperature differential is fixed at 1°C.

If, at the start of a defrost cycle, the temperature measured by the defrost probe is greater than the set end defrost value, the cycle is not performed.

In the event of defrost probe failure, the controller performs a timed defrost with a duration equal to the value set for dP. The same is true if the end defrost set point can not be reached, the defrost is stopped after a maximum time equal to the value of dP, and the error Ed is displayed (if enabled by A8) and stays on until a correct defrost cycle is performed, that is, one which ends by temperature.

dP Maximum defrost duration

Determines the maximum duration of the defrost (in minutes or seconds, see parameter dC). When defrost by time is set [d0= 2/3/4] this parameter represents the effective duration of the defrost.

d8 Alarm bypass time after defrost

Indicates the time the high temperature alarm is bypassed from the end of the defrost or from when the cold room door is opened, if the digital input is connected to the "door switch" (parameter A4).

AL absolute or relative temperature for low temperature alarm

Used to determine the activation threshold for the low temperature alarm.

AH absolute or relative temperature for high temperature alarm

Used to determine the activation threshold for the low temperature alarm.

Note: the low and high temperature alarms are automatically reset (this means that if the temperature returns within the allowed range, the alarm signal is automatically cancelled). During a high or low temperature alarm, the message 'HI' or 'LO', respectively, flashes on the display alternating with the temperature; if fitted, the buzzer is activated.

Ad temperature alarm delay

Indicates after how many minutes the temperature alarm is signalled from when it is detected. If the temperature, after the delay Ad, is back within the allowed limits, the alarm is not signalled.

A4: 3rd input configuration

The MULTIFUNCTION digital input can assume different meanings according to the value attributed to this parameter, and according to the model of controller used. Following is a description of the possible functions.

A4	meaning
7	Door switch with evaporator fans off 'dor' alarm: - instant (A7= 0) or delayed (A7>0) - operation of AUX output (H1= 3) Input open= door open Input closed= door closed

A7 Digital input alarm delay

Sets the delay (in minutes) for detecting the external alarm when A4 = 1, 7 or 8.

F0 Enable evaporator fan control

F0= 0 F0= 0 the fans are on;

F0= 1 the fans are controlled by the fan controller (see parameter F1). If the evaporator probe (2) is faulty, the fans are on.

F2 stop evaporator fans if compressor stops

F2= 0 the fans operate based on F0 even when the compressor is OFF;

F2= 1 the fans are OFF when the compressor is OFF.

F3 evaporator fan status during defrost

F3= 0 the fans are on during the defrost;

F3= 1 the fans are off during the defrost.

The device features 4 limited sets of parameters that can be pre-defined and used for the normal operation of the controller. This function is useful for the rapid configuration of the controller in which the value of just a few parameters needs to be changed.

Multiple set parameters:

code	parameter	type	min	max	UOM	default
EZY	Select set of default parameters	C	0	4	-	0

Table 4.n

Description of the parameter

EZY select the set of parameters

The "Easy Set" parameter is used to select one of the 4 quick configurations saved in the instrument, containing a maximum of 20 parameters each.

These parameters may vary between sets.

These parameters are the same for all sets, have default values sent with the instrument that the user can customise with the configuration key (programmed via pJEZ commissioning tool on a PC). Their value can be changed only via the key as they are saved to E2PROM.

If EZY is equal to 0, the default instrument parameters are used, and the values of all the parameters are visible and can be set from by the instrument (except for the masked parameters).

When a set is selected (EZY= 1 to 4), this set is loaded as default (0) and the visibility of the parameters does not change.

Any changes made to the parameters are saved to set 0, default, but not to the initial set 1 to 4 used; this means that if another set is loaded (including set 0), when returning to the previous set the modified values are not saved.

Sets 1 to 4 cannot contain the EZY parameter: if this occurs, when the set is loaded this is not transferred to set 0. This is required for security reasons, for the correct operation of the device.

H2 Enable keypad

Parameter 'H2' can be used to disable the setting of the set point and the other operating parameters when the instrument is located a place that is accessible to the public.

Specifically:

H2	Description
0	keypad disabled: - F parameters read-only - C parameters modifiable (settable PS) - no set point modification - no manual defrost - no continuous cycle - no ON/OFF

1	keypad enabled
2	keypad enabled except for ON/OFF

3.3 Table of Alarms

When an alarm is activated, the display shows the corresponding message (as illustrated in the table) that flashes alternating with the temperature; if fitted and enabled, the buzzer is also activated.

All the alarms have automatic reset (that is, they stop when the causes are no longer present), except for alarm 'CHt' which has manual reset (instrument on/off using the UP button).

alarm code	buzzer and alarm relay	LED	alarm description	parameters involved ENABLE ALARM
E0	active	ON	probe 1 error= control	-
E1	not active	ON	probe 2 error= defrost	d0 = 0 / 1 / 4
dOr	active	ON	open door alarm	[A4 = 7/8][+A7]
LO	active	ON	low temperature alarm	[AL] [Ad]
HI	active	ON	high temperature alarm	[AH] [Ad]
EE	not active	ON	unit parameter error	-
EF	not active	ON	operating parameter error	-
EtC	not active	ON	clock alarm	if bands active

4. Additional Info

- Because of possible higher humidity inside the appliance the electronic board is coated
- Additional info (manual, technical leaflet,...) can also be found on the Carel-homepage, follow this link:

[Homepage Carel easy](#)

APPENDIX A – TABLE OF PARAMETERS STATUS 27th of January 2011

Table of Parameters Carel easy controller 6114 553-00									
Type: F ...frequent (changeable by customer) / C ...customized (changeable with password) / H ...hidden (not changeable)									
#	Parameter	Description	Type	Unit	Main Set	Set 1	Set 2	Set 3	Set 4
1	PS	Password	H	-	22				
2	/2	Probe measurement stability	H	-	10				
3	/4	Select display probe	H	-	1				
4	/5	Select °C/°F	H	-	0				
5	/6	Disable decimal point	H	-	0				
6	/C1	Probe 1 offset	H	°C	-1,5				
7	/C2	Probe 2 offset	H	°C	0				
8	/C3	Probe 3 offset	H	°C	0				
9	St	User set point	H	°C	2				
10	r1	Minimum set point allowed	H	°C	2				
11	r2	Maximum set point allowed	H	°C	6				
12	r3	Direct / reverse	H	-	0				
13	r4	Night-time set point delta	H	°C	2				
14	rd	Differential	H	°C	2				
15	c0	Start delay on power-on	H	min	3				
16	c1	Delay between 2 consecutive starts	H	min	0				
17	c2	Minimum compressor OFF time	H	min	5				
18	c3	Minimum compressor ON time	H	min	2				
19	c4	Duty setting time	H	min	7				
20	cc	Continuous cycle duration	H	h	0				
21	c6	Temperature alarm bypass after cc	H	h	2				
22	d0	Type of defrost	H	-	0				

#	Parameter	Description	Type	Unit	Main Set	Set 1	Set 2	Set 3	Set 4
23	dI	Defrost interval	H	h/min see dC	9				
24	dt	Ende defrost temperature	H	°C	4,5				
25	dP	Maximum defrost duration	H	min/s see dC	120				
26	d4	Defrost at instrument power-on	H	-	0				
27	d5	Defrost delay from power-on	H	min	0				
28	d6	Display frozen during defrost	H	-	1				
29	dd	Dripping time	H	min	0				
30	d8	Alarm bypass after defrost	H	h	2				
31	d9	Defrost priority over compressor prot.	H	-	0				
32	d/	Defrost probe reading	H	°C	-				
33	dC	Time base	H	-	0				
34	A0	Alarms and fan differential	H	°C	1				
35	AL	Low temperature alarm	H	°C	2				
36	AH	High temperature alarm	H	°C	8				
37	Ad	Temperature alarm delay	H	min	180				
38	A4	3rd input configuration	H	-	7				
39	A7	Digital input alarm delay	H	min	2				
40	A8	Enable alarm 'Ed'	H	-	0				
41	Ac	Dirty cond. alarm setpoint	H	°C	70				
42	AE	Dirty cond. Alarm temp. Diff	H	°C	5				
43	Acd	Dirty cond. Alarm delay	H	min	0				
44	F0	Enable evaporator fan control	H	-	0				
45	F1	Evaporator fan control setpoint	H	°C	16				
46	F2	Stop evaporator fans if compr. Stops	H	-	0				
47	F3	Stop evaporator fans during defrost	H	-	0				
48	Fd	Post-dripping timeout	H	min	0				
49	H0	Supervisor serial address	H	-	1				

#	Parameter	Description	Type	Unit	Main Set	Set 1	Set 2	Set 3	Set 4
50	H1	AUX output configuration	H	-	0				
51	H2	Enable keypad	C	-	0				
52	H4	Disable buzzer	H	-	0				
53	H5	ID code (read only)	H	-	-				
54	EZY	Select set of parameters	H	-	0				
55	tEn	Enable RTC	H	-	0				
56	dAY	RTC day of the week	H	days	0				
57	hr	RTC hour	H	h	0				
58	Min	RTC minutes	H	min	0				
59	d1d	Defrost time band 1st day	H	days	0				
60	d1h	Time band 1st hour	H	h	0				
61	d1m	Time band 1st minutes	H	min	0				
62	d2d	Defrost time band 2nd day	H	days	0				
63	d2h	Time band 2nd hour	H	h	0				
64	d2m	Time band 2nd minutes	H	min	0				
65	d3d	Defrost time band 3rd day	H	days	0				
66	d3h	Time band 3rd hour	H	h	0				
67	d3m	Time band 3rd minutes	H	min	0				
68	d4d	Defrost time band 4th day	H	days	0				
69	d4h	Time band 4th hour	H	h	0				
70	d4m	Time band 4th minutes	H	min	0				
71	n0d	Night time band ON day	H	days	0				
72	n0h	Night time band ON hour	H	h	0				
73	n0m	Night time band ON minutes	H	min	0				
74	nFd	Night time band OFF day	H	days	0				
75	nFh	Nigh time band OFF hour	H	h	0				

#	Parameter	Description	Type	Unit	Main Set	Set 1	Set 2	Set 3	Set 4
76	nFm	Night time band OFF minutes	H	min	0				
77	A0d	AUX time band ON day	H	days	0				
78	A0h	AUX time band ON hour	H	h	0				
79	A0m	AUX time band ON minutes	H	min	0				
80	Afd	AUX time band OFF day	H	days	0				
81	AFH	AUX time band OFF hour	H	h	0				
82	Afm	AUX time band OFF minutes	H	min	0				

Waldner René
27^h of January, 2011